



Evan Dufraisie

NLP Research Engineer
PhD



Paris, France



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EvanDufraisie



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My LinkedIn



My Google Scholar



EvanD

Languages

French - (Native)

English - (C1-C2)

Spanish - (B2)

Experience

2023 - Present NLP Research Engineer

CEA List - Palaiseau

I took up the position in December 2023 while pursuing my Ph.D. I have been involved in several key responsibilities, including defining scientific roadmaps, managing internships, recruiting research engineers and prospective Ph.D. students, and serving as the IT correspondent. I'm currently involved in two major projects:

- Project manager and developer for an ongoing 7B parameter model pre-training on the Jean-Zay national cluster (training on 512 H100). Include also the development of data curation pipelines.
- Research engineer on a civil security project focused on RAG and LLM specialization: fine-tuning Mistral-7b models, synthetic-data generation, guided generation, tgi-inference deployment on cluster.

2021 - 2022 Teaching Assistant

Centrale Supélec - Gif-sur-Yvette

I animated the Algorithms and Complexity course for first year engineering students for two semesters.

2020 - 7 mo Research Internship

Talend & University of Nantes

Research at the Nantes university laboratory on the applicability of Bayesian Networks for anomaly detection and automatic correction in tabular data.

Education

2021 - 2024 Ph.D. in Computer Science

University of Lorraine - Nancy | CEA List - Palaiseau

My thesis is part of the ANR BOOM project, which focuses on combating polarization by developing innovative recommendation algorithms for press content using large deep language models. I have worked extensively on curating large amounts of data from CommonCrawl, domain adaptation, and fine-tuning BERT-based models for tasks such as Named Entity Recognition (NER) and Sentiment Analysis. This rich experience allowed me to collaborate with a multidisciplinary team on an impactful research initiative.

2019 - 2020 Master of Computer Science

The Georgia Institute of Technology - Atlanta (US)

Master of Computer Science with a specialization in Machine Learning. The coursework included advanced artificial intelligence, machine learning, deep learning, reinforcement learning, computer vision, algorithm design, and data visualization, with applications-focused courses in areas like finance and robotics.

2017 - 2020 Master of Computer Science

IMT-Atlantique (Mines Nantes) - Nantes

Master of Engineering with a specialization in Operations Research. The coursework included programming, software development, artificial intelligence, graph theory, advanced mathematics, numerical calculus, physics, and sociology.

2015 - 2017 Classes Préparatoires - PCSI-PC*

Lycée Georges Clémenceau - Nantes

Two-year intensive courses in mathematics, physics and chemistry for the French "Grandes Ecoles" entrance exams.

Volunteering

- 2022 - 3 mo **Season 10 - Data For Good** *Paris*
Worked on the Ceebios project, which aims to simplify biology papers using machine learning to make science open and accessible to everyone. I contributed as an advisor with my expertise in NLP and built a model for recognizing graph types in pdf.
- 2021 - 3 mo **Season 9 - Data For Good** *Paris*
Worked on the Open Fake Facts project, which aimed to detect fake news online. I introduced the team to advanced NLP techniques, helped with data scraping, and trained deep learning models.
- 2019 - 1yr **Mentoring in Classe Préparatoire** *Lycée Georges Clémenceau - Nantes*
Helped for a year a preparatory class student with physics and mathematics.

Software Stack

A non-exhaustive overview of some of the languages, tools, and environments I've used or use on a regular basis:

Programming Languages

Python | C++ | Java | Latex | notions of others

Libraries

Pytorch | Pytorch Lightning | DeepSpeed | Huggingface | Numba

Tools and Environment

Debian | Tmux | Vim | Git | Slurm | VsCode | bash

Publications

- 2023 **MAD-TSC: A Multilingual Aligned News Dataset for Target-dependent Sentiment Classification** *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics - (ACL)*
E Dufraisse, A Popescu, J Tourille, A Brun, J Deshayes
- 2022 **Being diverse is not enough: Rethinking diversity evaluation to meet challenges of news recommender systems** *Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization - (UMAP)*
C Treuillier, S Castagnos, E Dufraisse, A Brun
- 2022 **Don't burst blindly: for a better use of natural language processing to fight opinion bubbles in news recommendations** *Proceedings of the LREC 2022 workshop on Natural Language Processing for Political Sciences - (LREC)*
E Dufraisse, C Treuillier, A Brun, J Tourille, S Castagnos, A Popescu
- 2020 **Interactive Anomaly Detection in Mixed Tabular Data using Bayesian Networks** *Proceedings of the 10th International Conference on Probabilistic Graphical Models - (PMLR)*
E Dufraisse, P Leray, R Nedellec, T Benkhelif